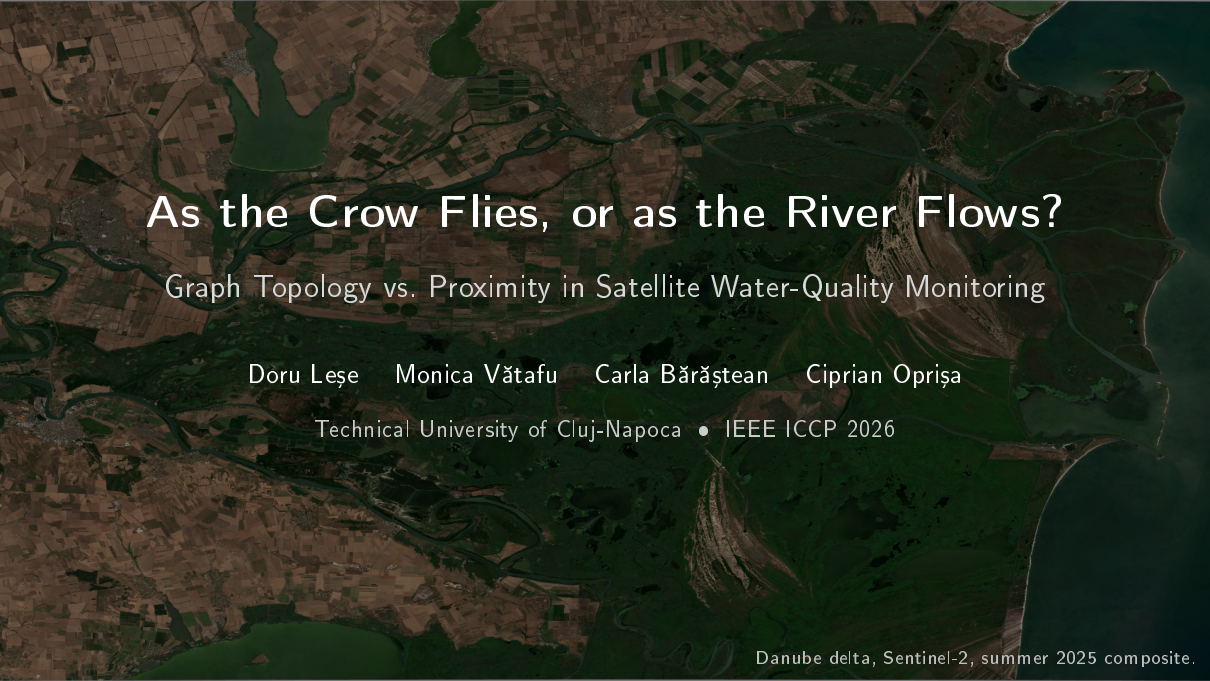
# As the Crow Flies, or as the River Flows?

## Graph Topology vs. Proximity in Satellite Water-Quality Monitoring

Doru Leșe  Monica Vătafu  Carla Bărăștean  Ciprian Oprișa

Technical University of Cluj-Napoca • IEEE ICCP 2026



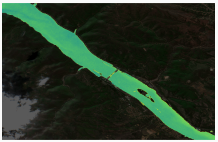
Danube delta, Sentinel-2, summer 2025 composite.

# Sentinel-2 data: surface reflectance and derived spectral indices



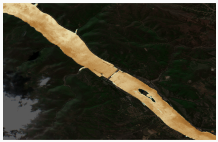
true colour

RGB (B4, B3, B2)



**NDCI**

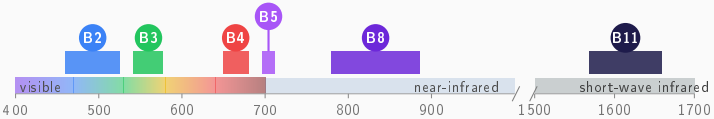
chl-a proxy (B5, B4)



**turbidity**

red reflectance (B4)

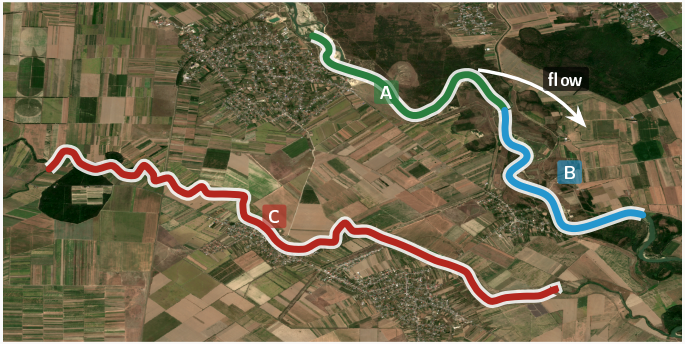
- Sentinel-2 L2A: 10/**20**/60 m resolution, ∼5-day revisit
- an *index* is a normalised band ratio, one value per pixel
- 7 indices
- indices are optical proxies, not concentrations




turbidity $= \mathrm{B4}$ NDCI $= (\mathrm{B5} - \mathrm{B4}) / (\mathrm{B5} + \mathrm{B4})$

Wavelength in nm. Bands drawn at their spectral width (Sentinel-2A). Images: Danube, Iron Gates, 2025-04-07.

# Water moves along the river, not in straight lines




The Siret and a tributary, eastern Romania, Sentinel-2, 2025-08-14.

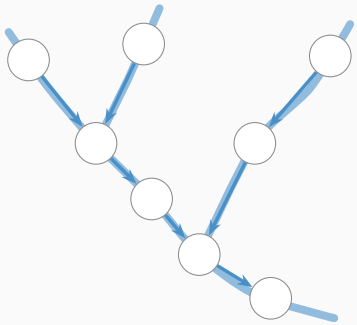
**A** and **B**: two consecutive stretches of the same river. Water from A reaches B.

**C**: a stretch of a tributary that A's water never enters. In a straight line, A to C is 5.2 km, A to B is 4.6 km.

C receives no flow from A.

Is this difference detectable in Sentinel-2 data?

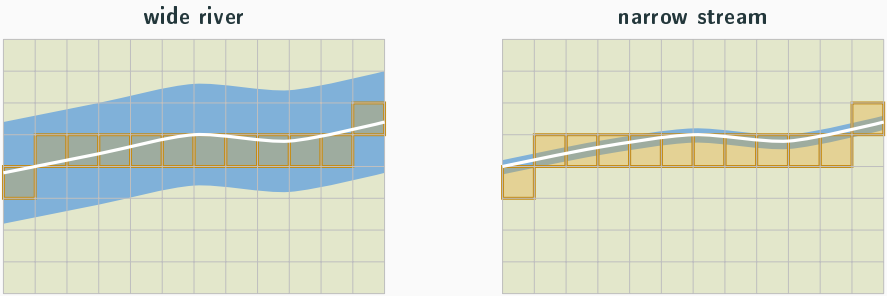
# A river network is a directed graph



- source: **EU-Hydro**, the public European river map of the Copernicus programme
- every river is a chain of short pieces: each **node** is one *reach*, typically about 2 km (median 1.7 km in Germany, 2.0 km in Romania)
- each reach records the reach just downstream of it: that is the **edge**
- acyclic: one downstream neighbour, or none where the mapped network ends

**flow neighbours** of a reach = the reaches one arrow away

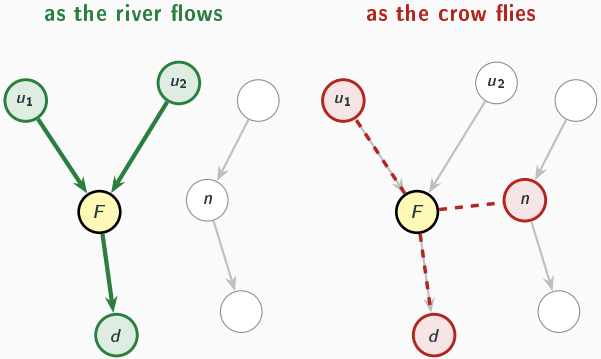
# From pixels to one number per reach




We average the 20 m pixels on the reach line, after per-pixel cloud masking: one number per reach per pass, then one per month.

On narrow streams, pixels mix water and bank, so values partly reflect land cover. We return to this.

# Two ways to choose a reach's neighbours



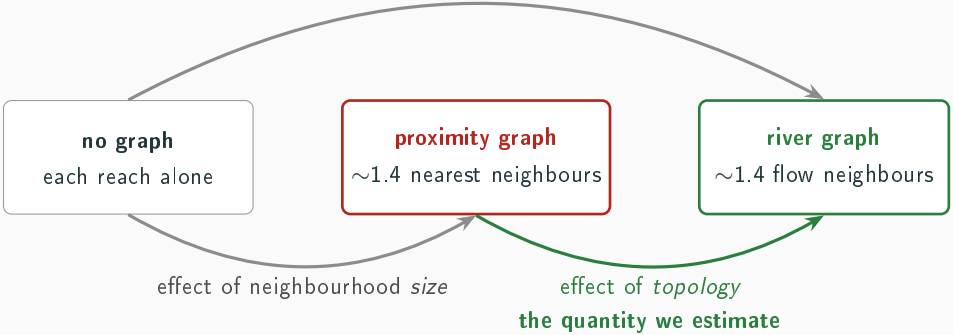

**River graph**: neighbours are flow-connected to $F$ (upstream $u_1, u_2$, downstream $d$).

**Proximity graph**: neighbours are simply the nearest reaches on the map. It swaps $u_2$ for $n$, which is close but not flow-connected.

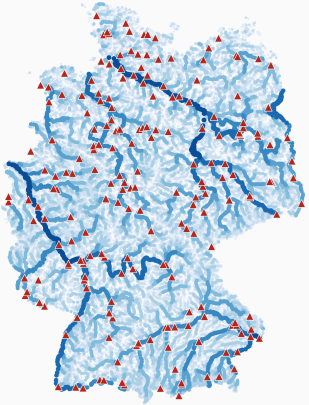
Both graphs have the same mean degree (about 1.4 on the monitored reaches). Only the neighbour sets differ.

# A controlled comparison: river graph vs. degree-matched proximity graph

common comparison in the literature [Shchur et al. 2018; Zhu et al. 2020]:
confounds *neighbourhood size* with *topology*



# Two national river graphs



Germany

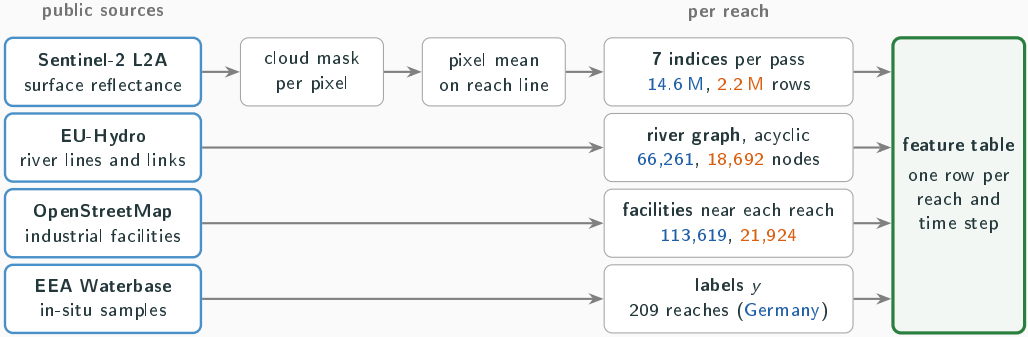
| | Germany | Romania |
|---|---|---|
| reaches | 66,261 | 18,692 |
| period | 2020 to 2023 | 07/2024 to 05/2026 |
| months | 48 | 23 |
| Sentinel-2 obs. | 14.6 M | 2.2 M |
| in-situ data* | **209** reaches (red) | none public |

*overlapping the Sentinel-2 era



Romania

# Processing pipeline: from public sources to one feature table




**The Leine** south of Göttingen

(RL26030359)

**Graph node**: Strahler 3, 1,723 m, 2 inflows

**Facilities**: 3 quarries within 2 km

**In-situ**: nitrate 28.3, 31.7 mg/L $\Rightarrow y{=}1$

**Sentinel-2 passes**, 7 indices (3 of 7 cloud-free passes in the month)

| date | NDWI | MNDWI | NDVI | NDTI | NDCI | BSI | turb. |
|---|---|---|---|---|---|---|---|
| 2022-08-02 | −0.716 | −0.538 | 0.770 | −0.121 | 0.413 | −0.233 | 458 |
| 2022-08-07 | −0.692 | −0.564 | 0.735 | −0.106 | 0.365 | −0.178 | 457 |
| 2022-08-12 | −0.683 | −0.568 | 0.718 | −0.083 | 0.355 | −0.147 | 472 |

# Labels: monthly exceedance of water-quality limits

### 1. Limits (LAWA good/moderate boundary)

| | |
|---|---|
| dissolved oxygen | below 6 mg/L |
| oxygen demand (BOD) | above 6 mg/L |
| ammonium | above 0.4 mg/L |
| nitrite | above 0.1 mg/L |
| nitrate | above 25 mg/L |
| chlorophyll-a | above 25 µg/L |

### 2. Rule

$y=1$ if **any** reading in the reach-month exceeds its limit, else $y=0$.

The label depends on sampling frequency.

### 3. Example: the Leine, two Augusts

| | | |
|---|---|---|
| 2022-08 | nitrate 28.3 and 31.7 **over 25**, 5 other readings fine | $y=1$ |
| 2023-08 | 3 readings, all fine | $y=0$ |

### 4. Resulting labels

46,154 readings ⇒ 8,100 reach-months. 126 of the 209 reaches exceed at least once.

**12.3%** $y=1$ (994) 87.7% $y=0$ (7,106)



### 5. Why detection will be weak

Most exceedances involve **dissolved**, optically inactive substances (ammonium, nitrate, oxygen). Chlorophyll-a exceeds its limit in only 16 reach-months.

# Four tests: what does the river graph add?

| Test | Question | Result |
|---|---|---|
| **1** Detection | Can satellite features flag a month that exceeds a limit? Does the river graph help? | next slides |
| **2** Co-fluctuation | Do flow-connected reaches move together more than equally close unconnected ones? | next slides |
| **3** Noise filter | Do river neighbours remove more false alarms than map neighbours? (synthetic events) | yes, rejects 26 / 36% vs. 2 / 13%, but by design: a sanity check |
| **4** Attribution | Does an elevated upstream show where a turbidity exceedance came from? | no: an unconnected control is elevated as often (83 vs. 86%, 80 vs. 78%) |

# Test 1 input: one sample per reach-month, 22 features

Example: the Leine south of Göttingen, the same two Augusts.

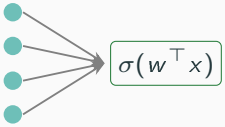
| | feature | 2022-08 | 2023-08 |
|---|---|---|---|
| satellite | NDWI (water) | −0.68 | −0.63 |
| | MNDWI (water) | −0.55 | −0.42 |
| | NDVI (vegetation) | 0.72 | 0.70 |
| | NDCI (chl-a proxy) | 0.358 | 0.357 |
| | NDTI (turbidity) | −0.10 | −0.16 |
| | turbidity | 499 | 547 |
| | BSI (soil index) | −0.17 | −0.25 |
| shape | Strahler order (river size) | 3 | 3 |
| | log reach length | 7.45 | 7.45 |
| | inflowing reaches | 2 | 2 |
| | confluence (0/1) | 1 | 1 |

| | feature | 2022-08 | 2023-08 |
|---|---|---|---|
| upstream | inflow turbidity, flow-weighted | 497 | 637 |
| | added here (if above inflow) | 2 | 0 |
| | value passed downstream | 499 | 547 |
| | discharge proxy (channel size) | 1.39 | 1.39 |
| | observed minus inflow | 2 | −90 |
| | mean of upstream reaches | 497 | 637 |
| facility | wastewater plants, 2 km | 0 | 0 |
| | industrial sites, 2 km | 0 | 0 |
| | factories, 2 km | 0 | 0 |
| | all facilities, 2 km | 3 | 3 |
| | nearest facility (m) | 1,242 | 1,242 |
| | **label** $y$ | **1** | **0** |

Near-identical optical features, opposite labels: the 2022 exceedance (nitrate) is optically inactive.

# Test 1 models: from logistic regression to a graph attention network

**Linear**



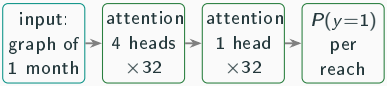

**logistic regression**
class-weighted, L2 penalty

**Tree ensembles**



**random forest**: 300 trees, averaged
**XGBoost**: gradient-boosted trees, 300 trees of depth 4, learning rate 0.05

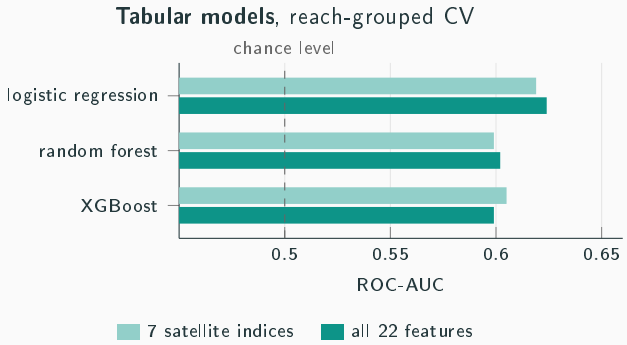
**Neural**

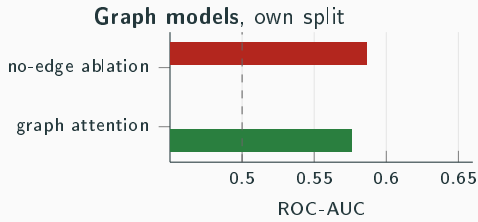



**graph attention network**: aggregates features of river-graph neighbours
**no-edge ablation**: same network, all edges removed

| | |
|---|---|
| **Input** | linear and trees: the 7 indices, then all 22 features. Graph models: 18 per reach. z-scored where needed |
| **Graph** | one per month: the 209 labelled reaches plus all reaches up to 5 steps upstream: 2,597 reaches |
| **Output** | predicted $P(y = 1)$ per reach-month |
| **Test** | 5-fold CV grouped by reach. Graph models are transductive: test features visible, labels masked |

# Test 1 results: weak signal, no significant gain from the river graph



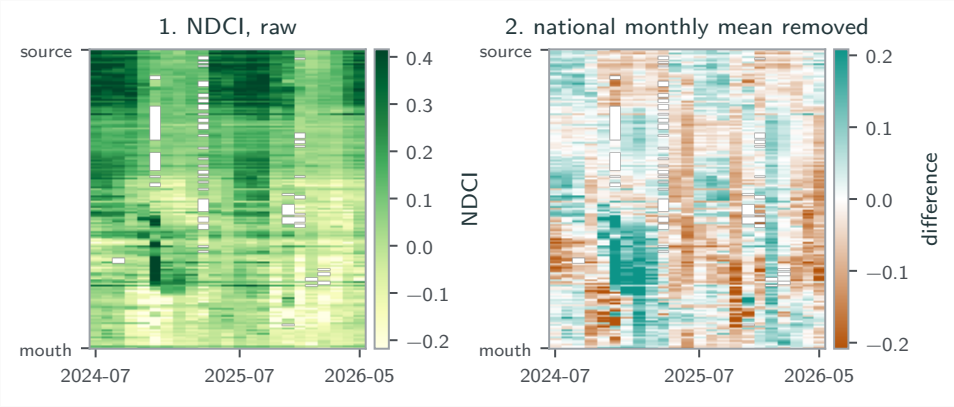

0.576 vs 0.586, 18 features per reach, 5 seeds.

**Weak, above chance.** ROC-AUC about 0.6 (chance 0.5). Average precision 0.18 (chance 0.13).

**Ranking lift.** Top 10% of reach-months: 23.5% exceed a limit, against a 12.6% base rate.

**No significant gain from the river graph.** Upstream features: average precision gain +0.002, 95% CI $[-0.019, +0.021]$. Graph attention below its ablation.

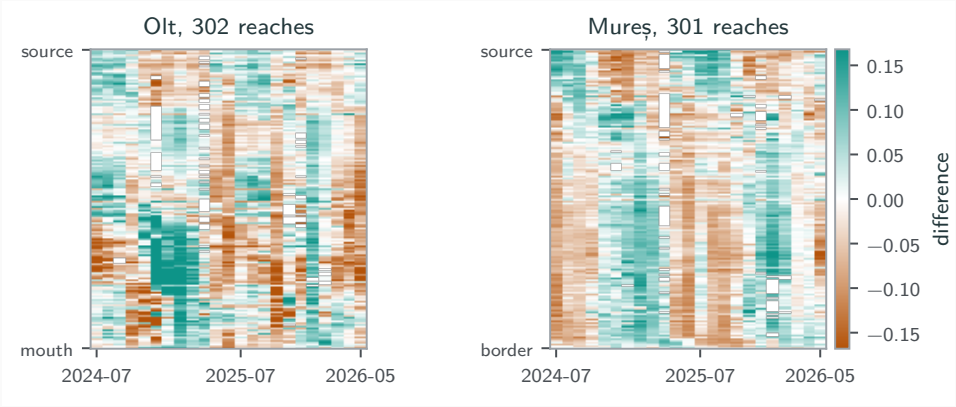
# Test 2: do flow-connected reaches co-fluctuate? Example: the Olt




Romania, monthly NDCI, 302 reaches from source to mouth. Right: each reach also centred on its own mean.

Left: a shared seasonal cycle dominates all reaches. Right: after removing the national monthly mean, contiguous stretches co-fluctuate, visible as vertical bands along the river.

# Test 2 example: two neighbouring rivers, Olt and Mureș

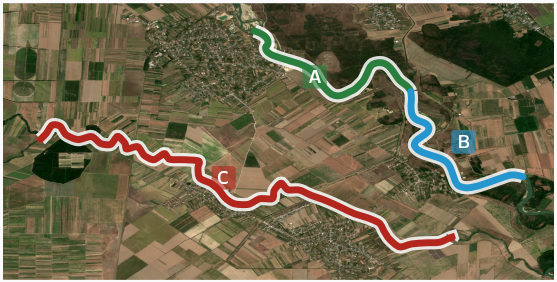



*"Oltule, râu blestemat, / Te-ai făcut adânc și lat,*
*Că vii mare, spumegat, / Și cu sânge-amestecat"*

*"Mureș, Mureș, apă lină,*
*Trece-mă-n țară străină / Și-mi fă parte de hodină"*

Sources 15 km apart: the Olt flows south, the Mureș west, each with its own bands.

# Test 2 design: a flow-connected pair vs. a distance-matched control




One of the 1,413 Romanian triples, Sentinel-2, 2025-08-14.

For a reach **A** we pick:

- **B**: the first reach downstream with data, at most 60 km along the river
- **C**: a reach *not* flow-connected to A, at about the same straight-line distance from A as B (0.7 to 1.3×)

Straight-line distance is matched. River size is not.

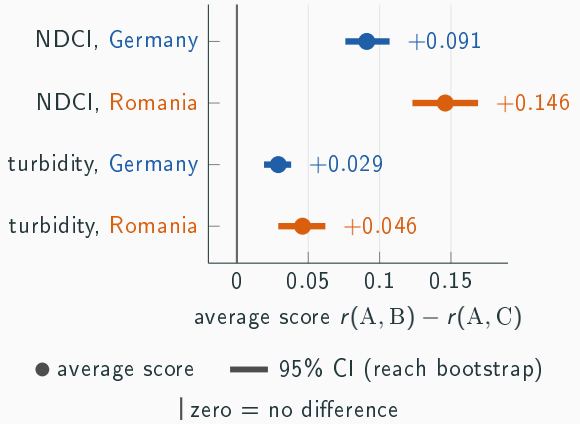
**Paired score** for reach A

$$r(\mathrm{A},\mathrm{B}) \;-\; r(\mathrm{A},\mathrm{C})$$

$$r(\mathrm{A},\mathrm{B}) = \frac{\sum_t (a_t - \bar{a})(b_t - \bar{b})}{\sqrt{\sum_t (a_t - \bar{a})^2}\,\sqrt{\sum_t (b_t - \bar{b})^2}}$$

Pearson correlation of the monthly series $a_t$, $b_t$ (national mean removed), over at least 12 shared months

# Test 2 result: flow-connected reaches co-fluctuate more than controls




CI: reaches resampled with replacement 2,000 times, middle 95% of the averages.

**What it shows**

- all four 95% CIs exclude zero
- in both countries
- NDCI: larger effect
- turbidity: smaller effect

**Caveat**: C often has a lower Strahler order than B. With size-matched pairs only, the NDCI effect shrinks but its CI still excludes zero. For turbidity in Romania, the CI then includes zero.

**Negative control**: with the series shuffled between reaches, the effect disappears.

# Conclusion: topology helps dynamics, not statics

| Test | River graph vs. degree-matched proximity | Verdict |
|---|---|---|
| **Test 1** static levels | no measurable gain: proximity already captures regional similarity. A GAT does not change this | no advantage |
| **Test 2** dynamics | flow-connected reaches co-fluctuate more: $\Delta r \approx 0.03$ to 0.15, both indices, both countries | modest but robust |
| **Test 3** noise filtering | false alarms rejected: 26% / 36% vs. 2% / 13% | helps, synthetic only |
| **Test 4** attribution | upstream co-elevated no more often than an unconnected control | no advantage |

**In practice**: a guide to *where to sample first*, not a replacement for lab sampling. The top 10% of reach-months exceed a limit 1.9× more often (23.5% vs 12.6%).

**Before you follow the river, ask the crow.**

Compare any graph model with a degree-matched proximity graph.

Sometimes the river wins. Either way, the trip is worth it.

# Limitations and outlook

## Limitations

- **Clouds**: at 1-day bins about 85% of reach-days have no cloud-free pass, which limits temporal resolution
- **Labels**: in-situ chemistry only for Germany, none for Romania in the Sentinel-2 era
- **Optical proxies**: NDCI not validated against chlorophyll-a (84 matched reach-months)
- **Two networks, not independent**: same sensor, same hydrography, temperate climate, shared Danube drainage
- **Ground truth**: event tests rely on synthetic injections, partly circular

## Outlook

- Labels, not the method, bound the study: wider publication of monitoring data would relax this
- The protocol is portable: EU-Hydro covers all EU states, all inputs are public
- The graph is sensor-agnostic: higher-revisit satellites improve features without changes

Code, panels, per-pass archives, splits and seeds:

`github.com/ldcx1/river-flow-vs-proximity`



# Thank you.

Questions?

github.com/ldcx1/river-flow-vs-proximity

# References


**Data**

- Copernicus Land Monitoring Service, *EU-Hydro River Network Database*.
- European Space Agency, *Sentinel-2 MSI Level-2A surface reflectance*, via Google Earth Engine.
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**Remote sensing of rivers**

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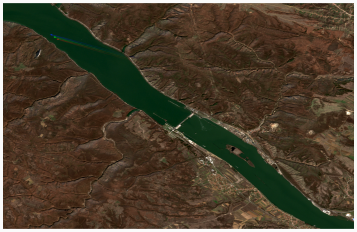
**Graphs and evaluation**

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- J. Zhu et al., Beyond homophily in graph neural networks, *NeurIPS*, 2020.
- P. Veličković et al., Graph attention networks, *ICLR*, 2018.
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**River networks**

- J. M. Ver Hoef, E. E. Peterson, Spatial statistical models of stream networks, *J. Am. Stat. Assoc.* 105, 2010.
- X. Jia et al., Physics-guided recurrent graph model for river networks, *SIAM SDM*, 2021.
- A. Sarkar et al., HydroGAT, *ACM SIGSPATIAL*, 2025.
- K. Buchhorn et al., GNN anomaly detection for river networks, *F1000Research* 12, 2023.

# Test 3: optical anomalies are detectable, but event labels are hard to obtain



2025-03-20

B4 reflectance: 2.2%



2025-04-07

B4 reflectance: 10.5%

Same reach, 18 days apart: red reflectance rose, consistent with more sediment.

No easy-to-parse public record says where and when events happened.

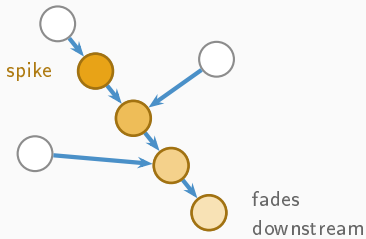
So we test detection with injected, synthetic events.

Danube at the Iron Gates, Sentinel-2. Turbidity proxy of this reach on each pass.

Land cover also changed: spring leaf-out between the two dates.

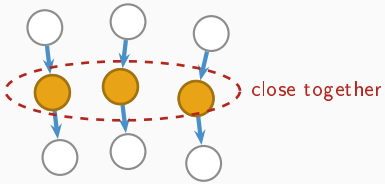
# Test 3 setup: two injected types, one filter rule

**plume = true event**




a spike at one reach,
carried 2 to 5 reaches
further downstream,
fading at each step

**cluster = false alarm**



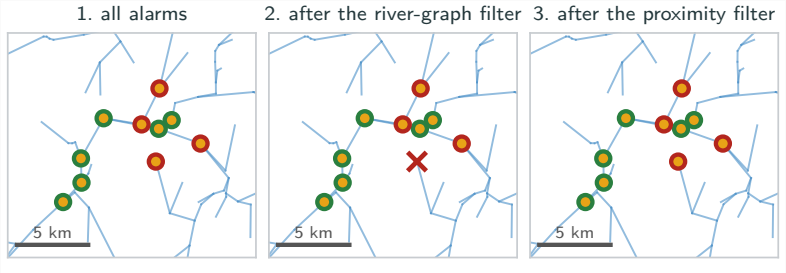

3 to 5 nearby reaches raised together,
with no water link between them

Detector: flag the reaches above the month's 98th percentile.

**Filter**: keep an alarm only if a neighbour also has one.

River graph: within 4 reaches up or downstream. Proximity: within 6 km.

# Test 3: with injected events, the river-graph filter rejects more false alarms




injected event  false alarm  × removed  river graph

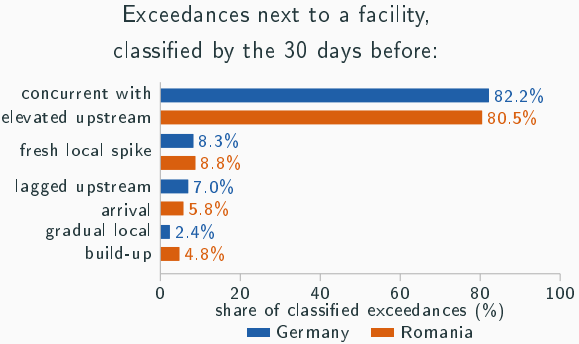
| 20 runs | false alarms removed | | true-event alarms kept | |
|---|---|---|---|---|
| | Germany | Romania | Germany | Romania |
| river-graph filter | **26%** | **36%** | 98% | 99% |
| proximity filter | 2% | 13% | 99.7% | 99.7% |

Injections follow the flow by construction: a sanity check, not a validation.

# Test 4: can a turbidity exceedance be attributed to a nearby facility?



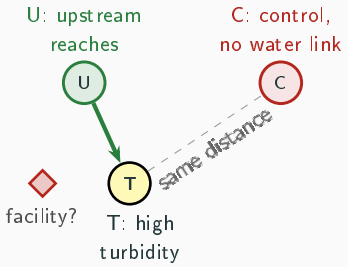

Setting only, no event claimed. Hârșova, Sentinel-2, 2025-07-25.




Most are **concurrent with an elevated upstream**: this suggests an upstream origin.

Is it flow transport, or a regional signal such as rainfall?

# Test 4: does high turbidity upstream show where it came from?




**High**: above national 95th percentile
**U**: up to 3 steps upstream of T
**C**: as many nearest, no water link
**Same time**: within ±6 days

**Idea** The high turbidity at T came down the river from U.

**If true** U is high at the same time more often than C, a control group just as close but not flow-connected.

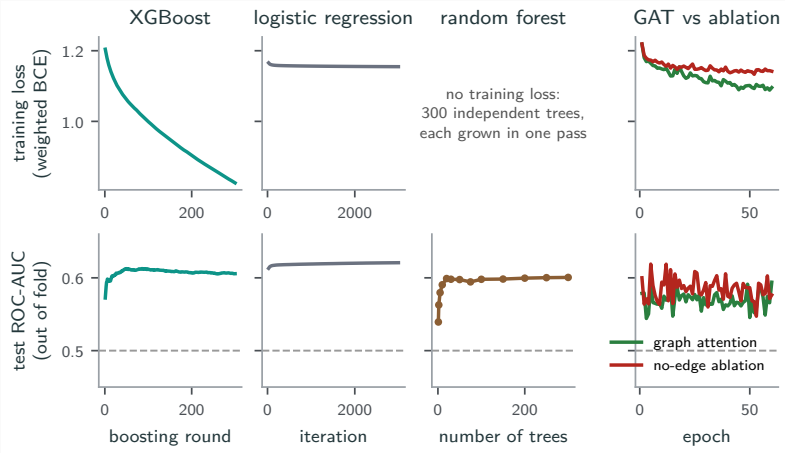
**Measured** Share of T exceedances with the group also high:

| | U upstream | C control | gap |
|---|---|---|---|
| Germany | 83.3% | 85.6% | −2.2 |
| Romania | 79.8% | 78.0% | +1.8 |

**Result** U is not meaningfully more often high than C. The gap is small and flips sign.

On average, co-elevation does not follow the flow link: upstream data alone cannot single out the source.

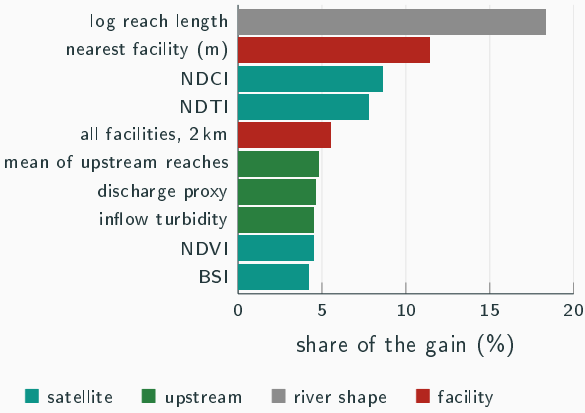
# Test 1: training curves




Loss falls, test ROC-AUC stays near 0.6: the limit is the signal, not the fitting. The forest is flat after 20 trees. GAT ahead of its no-edge ablation in only 8 of 60 epochs: the null result does not depend on the epoch.

Seed 0, paper settings, reach-grouped 5-fold CV, no early stopping. Last points match the logged seed-0 results.

# Test 1: what the trees split on




**Gain**: how much a feature's splits lower the training loss.

By group: satellite 35%, facility 27%, river shape 20%, upstream 19%.

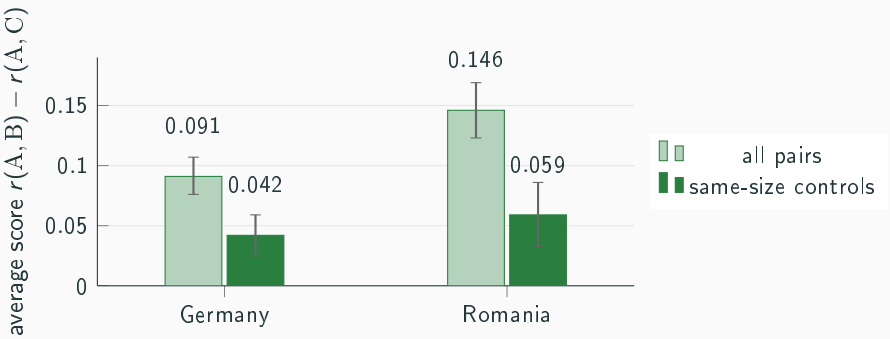
Length and facility distance never change over time: they let the trees *identify individual reaches*.

Under reach-grouped CV they add little: +0.01 ROC-AUC for river shape, none for facilities. Split gain does not imply predictive value.

Boosted trees with the paper's settings, fit once on all 7,739 reach-months and all 22 features.

# Robustness: is the co-fluctuation effect confounded by river size?

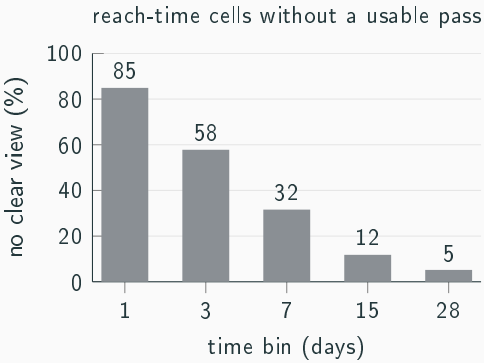
Controls are chosen by distance, not by size. Big rivers are rare, so a control is often a smaller stream.
**Check**: keep only pairs where C has about the same size as A (Strahler order within one).

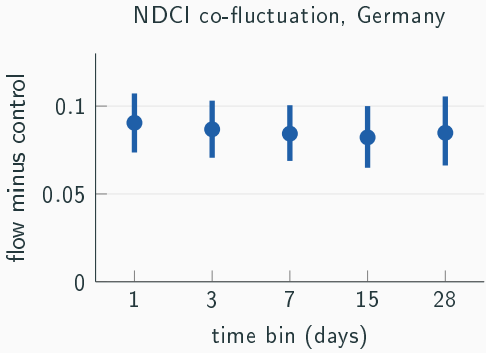



The gap shrinks, but it stays above zero in both countries.
The same holds with per-reach seasonal anomalies and with raw values.

NDCI, national monthly mean removed.

# From days to months: clouds set the limit




The effect is stable across bin widths, so it is not an artefact of monthly averaging. But at 1 day, 85% of cells have no usable image: no overpass that day, or clouds.

# All label-free experiments (paper Table II)

| Experiment | Metric | Germany | Romania |
|---|---|---|---|
| *Static levels: is a place similar to its neighbour?* | | | |
| turbidity (not distance-matched) | Spearman $\Delta r$ | no verdict $-0.098$ | tie $-0.010$ |
| NDCI (not distance-matched) | Spearman $\Delta r$ | tie $+0.019$ | win $+0.034$ |
| *Temporal dynamics* | | | |
| NDCI, country average removed | $\Delta$corr | win $+0.091$ | win $+0.146$ |
| turbidity, country average removed | $\Delta$corr | win $+0.029$ | win $+0.046$ |
| *Synthetic anomalies injected along the flow* | | | |
| recover a connected event | F1 | win $+0.019$ | win $+0.038$ |
| suppress false alarms | precision | win $+0.007$ | win $+0.018$ |
| flag a break between neighbours | ROC-AUC | win $+0.007$ | tie $+0.002$ |
| own vs inherited signal | ROC-AUC | tie $-0.002$ | tie $-0.001$ |
| *Scaling* | | | |
| co-fluctuation across basins | $\Delta$corr | win $+0.012$ to $+0.035$ | win $+0.029$ to $+0.068$ |

Effect = flow graph minus proximity control. win: 95% CI above zero, tie: CI includes zero. Intervals in the paper. Synthetic F1 levels are low (0.059 vs 0.040 in Germany), so only relative structure is meaningful.

# Model choice has little effect on Test 1 (paper Table I)

| | satellite only | | all features | |
|---|---|---|---|---|
| Model | ROC-AUC | avg. precision | ROC-AUC | avg. precision |
| logistic regression | 0.619 | 0.184 | 0.624 | 0.184 |
| random forest | 0.599 | 0.180 | 0.602 | 0.173 |
| XGBoost (reference) | 0.605 | 0.183 | 0.599 | 0.178 |
| MLP, no edges† | — | — | 0.586 | 0.161 |
| GAT, directed† | — | — | 0.576 | 0.151 |

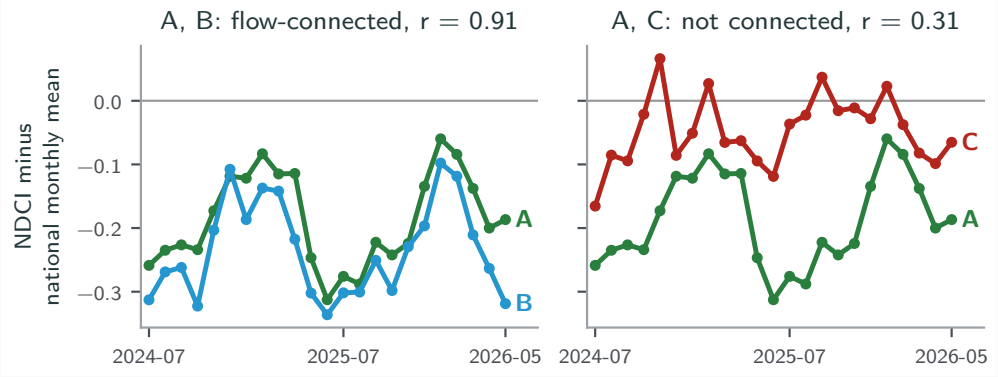
Five re-splits of 5-fold CV grouped by reach (no reach in both train and test). Average precision of a random order = 0.126. †Separate 2,597-node subgraph and split: only the GAT vs MLP gap is meaningful ($-0.010 \pm 0.019$ over five seeds). GAT: 2 layers, 32 hidden, 4 heads, 60 epochs, untuned.

# The two Romanian exceptions

- **static NDCI pairing**: flow beats proximity, $+0.034$ $[+0.011, +0.056]$
- **Romanian detector on coarse EU status labels** (2,268 water bodies, 2022): satellite only ROC-AUC 0.610, with upstream features 0.653 ($+0.043$ $[+0.026, +0.059]$)
- but the same upstream features *hurt* when a German model is transferred zero-shot ($-0.065$ $[-0.091, -0.038]$)

The coarse body-level label is not comparable with the German monthly labels, so neither exception overturns the static null result.

# Test 2 example: monthly NDCI of one triple




NDCI of A, B and C from the last slide, 23 months, national monthly mean removed.
A and B are strongly correlated. C is only weakly correlated with A.

Score for this A: $r(\mathrm{A},\mathrm{B}) - r(\mathrm{A},\mathrm{C}) = 0.91 - 0.31 = +0.60$.